

A Comprehensive Analysis of 120 Years of Olympic Games Data Using SQL (1896–2024)

Dataset: <https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results>

Q1. How many Olympics games have been held?

```
1
2  -- Questions
3  -- Q1. How many Olympics games have been held.
4  • SELECT COUNT(DISTINCT Games) AS total_games
5  FROM athlete_event;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
total_games			
51			

Q2. List down all Olympics Games held so far.

```
6  -- Q2. List down all Olympics Games held so far.
7  • SELECT DISTINCT Games
8  FROM athlete_event;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
Games				
1992 Summer				
2016 Summer				
2012 Summer				
1920 Summer				
1900 Summer				
1988 Winter				
1992 Winter				
1994 Winter				
1952 Summer				
1960 Summer				

Q3. Mention the total number of nations that participated in each olympic games.

```

9      -- Q3. Mention the total number of nations that participated in each olympic
10 •   SELECT Games, COUNT(DISTINCT region) AS Total_participated_nations
11     FROM athlete_event a
12     JOIN noc_regions b
13     ON a.NOC = b.NOC
14     GROUP BY Games
15     ORDER BY Games;

```

Result Grid		
	Games	Total_participated_nations
▶	1896 Summer	12
	1900 Summer	31
	1904 Summer	14
	1906 Summer	20
	1908 Summer	22
	1912 Summer	29
	1920 Summer	29

Result 9 ×

Q4. which year saw the highest and lowest number of countries participating in the Olympics.

```

16      -- Q4. which year saw the highest and lowest number of countries participating in the olympics
17 •   (SELECT Year, COUNT(DISTINCT region) AS cnt
18     FROM athlete_event a
19     JOIN noc_regions b
20     ON a.noc = b.noc
21     GROUP BY Year
22     ORDER BY cnt
23     LIMIT 1)
24     UNION
25     (SELECT Year, COUNT(DISTINCT region) AS cnt
26     FROM athlete_event a
27     JOIN noc_regions b
28     ON a.noc = b.noc
29     GROUP BY Year
30     ORDER BY cnt DESC
31     LIMIT 1);

```

Result Grid		
	Year	cnt
▶	1896	12
	2016	204

```

32  -- Using window-function
33  • WITH cte AS (SELECT Year,
34      COUNT(DISTINCT region) AS cnt,
35      RANK()OVER(ORDER BY COUNT(DISTINCT region) DESC) AS participated_desc_rnk,
36      RANK()OVER(ORDER BY COUNT(DISTINCT region) ASC) AS participated_asc_rnk
37      FROM athlete_event a
38      JOIN noc_regions b
39      ON a.noc = b.noc
40      GROUP BY Year)
41  SELECT Year, cnt
42  FROM cte
43  WHERE participated_desc_rnk=1
44  OR participated_asc_rnk=1 ;

```

Result Grid		
	Filter Rows:	Export: Wrap Cell Content:
Year	cnt	
1896	12	
2016	204	

Q5. Which nation has participated in all of the Olympic Games?

```

45  -- Q5. Which nation has participated in all of the Olympic Games.
46  • SELECT region, COUNT(DISTINCT Games) FROM noc_regions a
47  JOIN athlete_event b
48  ON a.noc = b.noc
49  GROUP BY region
50  HAVING COUNT(DISTINCT Games) = (SELECT COUNT(DISTINCT Games) FROM athlete_event);

```

Result Grid		
	Filter Rows:	Export: Wrap Cell Content:
region	COUNT(DISTINCT Games)	
France	51	
Italy	51	
Switzerland	51	
UK	51	

Q6. Identify the sports played in all Summer Olympics?

```

51  -- Q6. Identify the sports played in all Summer olympics.
52  •  SELECT * FROM athlete_event;
53  •  SELECT sport
54      FROM athlete_event
55      WHERE Season = 'Summer'
56      GROUP BY sport
57      HAVING COUNT(DISTINCT Games) = (SELECT COUNT(DISTINCT Games) FROM athlete_event
58                                      WHERE Season = 'Summer'
59                                      );

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

sport
▶ Athletics
Cycling
Fencing
Gymnastics
Swimming

Q7. Which sports were played only once in the Olympics?

```

60  -- Q7. Which sports were played only once in the olympics.
61  •  SELECT sport
62      FROM athlete_event
63      GROUP BY sport
64      HAVING COUNT(DISTINCT Games) = 1;

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content:  | Fetch

sport
▶ Aeronautics
Basque Pelota
Cricket
Croquet
Jeu De Paume
Military Ski Patrol
Motorboating
Racquets
Roque
Rugby Sevens

Q8. Fetch the total number of sports played in each olympic game?

```

65  -- Q8. Fetch the total number of sports played in each olympic game.
66  •  SELECT Games, COUNT(DISTINCT Sport) as number_of_sports
67      FROM athlete_event
68      GROUP BY Games;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
Games	number_of_sports			
1896 Summer	9			
1900 Summer	20			
1904 Summer	18			
1906 Summer	13			
1908 Summer	24			
1912 Summer	17			
1920 Summer	25			
1924 Summer	20			
1924 Winter	10			
1928 Summer	17			

Q9. Fetch details of the oldest athletes to win the Gold medal?

```

69  -- Q9. Fetch details of the oldest athletes to win the Gold medal
70  -- using where clause
71  •  SELECT * FROM athlete_event
72      WHERE Medal = 'Gold'
73      ORDER BY Age DESC
74      LIMIT 1;

```

Result Grid		Filter Rows:		Edit:		Export/Import:		Wrap Cell Content:		Fetch rows:			
ID	Name	Sex	Age	Height	Weight	Team	NOC	Games	year	Season	City	Sport	Event
117046	Oscar Gomer Swahn	M	64	NULL	NULL	Sweden	SWE	1912 Summer	1912	Summer	Stockholm	Shooting	Shooting Men's Running
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```

75  -- using subquery
76  •  SELECT * FROM athlete_event
77      WHERE Medal = 'Gold'
78      AND Age = (SELECT MAX(Age) FROM athlete_event WHERE Medal = 'Gold');

```

Result Grid														Filter Rows:		Edit:		Export/Import:		Wrap Cell Content:	
	ID	Name	Sex	Age	Height	Weight	Team	NOC	Games	year	Season	City	Sport	Event							
▶	117046	Oscar Gomer Swahn	M	64	NULL	NULL	Sweden	SWE	1912 Summer	1912	Summer	Stockholm	Shooting	Shooting Men's Running							
	53238	Charles Jacobus	M	64	NULL	NULL	United States	USA	1904 Summer	1904	Summer	St. Louis	Roque	Roque Men's Singles							
▲	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL							

120 years of Olympics: A SQL study

```
79 -- Using window function
80 WITH cte AS (SELECT *,
81 RANK()OVER(ORDER BY Age DESC) AS rnk
82 FROM athlete_event
83 WHERE Medal = 'Gold')
84 SELECT *
85 FROM cte
86 WHERE rnk=1;
```

ID	Name	Sex	Age	Height	Weight	Team	NOC	Games	year	Season	City	Sport	Event
53238	Charles Jacobus	M	64	NULL	NULL	United States	USA	1904 Summer	1904	Summer	St. Louis	Roque	Roque Men's Singles
117046	Oscar Gomer Swahn	M	64	NULL	NULL	Sweden	SWE	1912 Summer	1912	Summer	Stockholm	Shooting	Shooting Men's Running

Q10. Find the ratio of Male and Female athletes participation in olympic Games?

```
87 -- Q10. Find the ratio of Male and Female athletes participation in olympic Games.
88 SELECT Games,
89 COALESCE(CONCAT('1 : ',
90 ROUND(COUNT(DISTINCT CASE WHEN Sex = 'M' THEN Name END)/
91 COUNT(DISTINCT CASE WHEN Sex = 'F' THEN Name END),0)), '1 : 0')AS female_male_ratio
92 FROM athlete_event
93 GROUP BY Games;
```

Games	female_male_ratio
1896 Summer	1 : 0
1900 Summer	1 : 52
1904 Summer	1 : 107
1906 Summer	1 : 139
1908 Summer	1 : 45
1912 Summer	1 : 44
1920 Summer	1 : 33
1924 Summer	1 : 20
1924 Winter	1 : 23
1928 Summer	1 : 9

Q11. Fetch the top 5 athletes who won the most Gold Medals

```
94  -- Q11. Fetch the top 5 athletes who won the most Gold Medals
95  • WITH CTE AS (SELECT ID, Name, row_number()OVER(PARTITION BY ID) AS rnk
96    FROM athlete_event
97    WHERE Medal = 'Gold')
98    SELECT Name, MAX(rnk)
99    FROM CTE
100   GROUP BY Name
101   ORDER BY MAX(rnk) DESC
102   LIMIT 5;
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	Name	MAX(rnk)
►	Michael Fred Phelps, II	23
	Raymond Clarence "Ray" Ewry	10
	Larysa Semenivna Latynina (Diriy-)	9
	Paavo Johannes Nurmi	9
	Frederick Carlton "Carl" Lewis	9

Q12. Fetch the top 5 athletes who have won the most medals (gold/silver/bronze).

```
103  -- Q12. Fetch the top 5 athletes who have won the most medals (gold/silver/bronze).
104  •  SELECT Name, COUNT(Medal) AS medal_cnt FROM athlete_event
105      WHERE Medal in ('Gold', 'Silver', 'Bronze')
106      GROUP BY Name
107      ORDER BY medal_cnt DESC
108      LIMIT 5;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
Name	medal_cnt			
Michael Fred Phelps, II	28			
Larysa Semenivna Latynina (Diriy-)	18			
Nikolay Yefimovich Andrianov	15			
Edoardo Mangiarotti	13			
Takashi Ono	13			

Q13. Fetch the top 5 most successful countries in the Olympics based on the number of medals won.

```
109  -- Q13. Fetch the top 5 most successful countries in the Olympics based on the number of medals won.
110  •  SELECT region, count(Medal) AS medal_cnts FROM athlete_event tb1
111      JOIN noc_regions tb2
112      ON tb1.NOC = tb2.NOC
113      WHERE Medal in ('Gold', 'Silver', 'Bronze')
114      GROUP BY region
115      ORDER BY medal_cnts DESC
116      LIMIT 5;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
region	medal_cnts			
USA	5637			
Russia	3947			
Germany	3756			
UK	2067			
France	1767			

Q14. List down total Gold, Silver and Bronze medals won by each country.

```

117 -- Q14. List down total Gold, Silver and Bronze medals won by each country.
118 • SELECT region, sum(Medal = 'Gold') as gold_med,
119      sum(Medal = 'Silver') as silver_med, sum(Medal = 'Bronze') as Bronze_med,
120      COUNT(Medal) as total_med FROM athlete_event tb1
121 JOIN noc_regions tb2
122 ON tb1.NOC = tb2.NOC
123 WHERE Medal in ('Gold', 'Silver', 'Bronze')
124 GROUP BY region
125 ORDER BY total_med DESC;

```

Result Grid					
	region	gold_med	silver_med	Bronze_med	total_med
▶	USA	2638	1641	1358	5637
	Russia	1599	1170	1178	3947
	Germany	1301	1195	1260	3756
	UK	677	739	651	2067
	France	499	602	666	1767
	Italy	575	531	531	1637
	Sweden	479	522	535	1536
	Canada	463	438	451	1352
	Australia	368	459	522	1349
	Hungary	432	332	371	1135

Q15. In which Sport/event did India win the highest number of medals?

```
126 -- Q15. In which Sport/event did India win the highest number of medals?
127 • SELECT Sport, COUNT(DISTINCT CONCAT(Games, Event, Medal)) AS medal_cnt
128 FROM saiqa.athlete_event as t1
129 JOIN noc_regions as t2
130 ON t1.NOC = t2.NOC
131 WHERE team = 'INDIA'
132 AND Medal <> 'NA'
133 GROUP BY Sport
134 ORDER BY medal_cnt DESC
135 LIMIT 1;
136
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	Sport	medal_cnt			
▶	Hockey	11			